

A Spectral-Synthesis-Free Katznelson–Tzafriri Theorem for Measures

A candidate full answer to Remark 1.3 of arXiv:1410.1301

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Abstract

Seifert asks whether the spectral-synthesis assumption can be removed from the Hilbert-space Katznelson–Tzafriri theorem for finite measures. We give an affirmative answer. The high-frequency resolvent hypothesis forces asymptotic uniform norm continuity. This supplies exactly the regularity needed to combine an L^1 -smoothed Hilbert ultrapower with the unitary asymptote and the spectral theorem, even when the original measure is singular.

Status. Candidate full result; verifier verdict: likely valid. No spectral-synthesis assumption remains. Human review is recommended, especially for the continuous-ultrapower and measure-integral steps.

1 Source question

The source is Seifert, *A Katznelson–Tzafriri theorem for measures*, arXiv:1410.1301 [1]. For a finite complex Borel measure μ on $\mathbb{R}_+$, it uses

$$(\mathcal{F}\mu)(s) = \int_{\mathbb{R}_+} e^{-ist} d\mu(t), \quad \widehat{\mu}(T)x = \int_{\mathbb{R}_+} T(t)x d\mu(t).$$

Theorem 1.2 assumes that T is a bounded C_0 -semigroup on a complex Hilbert space, with generator A , and that for some $R > 0$,

$$\{is : |s| \geq R\} \subset \rho(A), \quad \sup_{|s| \geq R} \|R(is, A)\| < \infty. \quad (1)$$

If $i\sigma(A) \cap \mathbb{R}$ is additionally a set of spectral synthesis and $\mathcal{F}\mu$ vanishes there, it concludes that $\|T(t)\widehat{\mu}(T)\| \rightarrow 0$.

Remark 1.3 on p. 2 asks:

It remains open whether [the spectral-synthesis assumption can be dropped] also in the setting of Theorem 1.2.

2 Proof intuition

For L^1 kernels, Seifert’s Hilbert-space proof smooths the functional calculus in variation norm before passing to an ultrapower [2]. That cannot be copied literally for a singular measure: convolution with an L^1 approximate identity does not converge to a singular measure in total variation.

space, the following generalisation of Theorem [1.1] was recently obtained in [6] Theorem 6.14].

Theorem 1.2. *Let X be a complex Hilbert space and let T be a bounded C_0 -semigroup on X with generator A . Suppose that $i\sigma(A) \cap \mathbb{R}$ is of spectral synthesis and that there exists $R > 0$ such that $\{is : |s| \geq R\} \subset \rho(A)$ and $\sup\{\|R(is, A)\| : |s| \geq R\} < \infty$. Then, given any bounded Borel measure μ on $\mathbb{R}_+$ such that $(\mathcal{F}\mu)(s) = 0$ for all $s \in i\sigma(A) \cap \mathbb{R}$, $\|T(t)\hat{\mu}(T)\| \rightarrow 0$ as $t \rightarrow \infty$.*

Here the Fourier transform $\mathcal{F}\mu$ and the operator $\hat{\mu}(T)$ are defined, for any bounded Borel measure on $\mathbb{R}$ and $\mathbb{R}_+$, respectively, and for any bounded C_0 -semigroup T , by formulas analogous to (1.1) and (1.2), with the measure $a(t)dt$ replaced by $d\mu(t)$ in each case. Thus Theorem [1.2] generalises Theorem [1.1] in the Hilbert space setting by no longer requiring the measure μ to be absolutely continuous with respect to Lebesgue measure.

Remark 1.3. It is shown in [22] Theorem 3.1] that the assumption of spectral synthesis can be dropped in Theorem [1.1] when X is a Hilbert space. It remains open whether this is true also in the setting of Theorem [1.2]

Figure 1: Theorem 1.2 and Remark 1.3 on p. 2 of arXiv:1410.1301.

The resolvent assumption provides the missing replacement. On Hilbert space it gives a negative critical growth bound, which implies that T becomes uniformly norm-continuous far out on the time axis. We therefore smooth the *almost-norming vectors* rather than the measure. The smoothing places those vectors in the continuous part of a Hilbert ultrapower, while tail norm-continuity preserves the non-decay witnessed by the original singular measure.

The unitary asymptote of that continuous ultrapower sees only the original boundary spectrum. On this unitary group the spectral theorem turns the measure calculus into multiplication by $\mathcal{F}\mu$, so the induced operator is zero. The resulting orbit must both decay and stay uniformly away from zero, which is the contradiction.

3 Main result

Theorem 1. *Let H be a complex Hilbert space and let $T = (T(t))_{t \geq 0}$ be a bounded C_0 -semigroup on H , with generator A , satisfying (1). Let μ be a finite complex Borel measure on $\mathbb{R}_+$ such that*

$$(\mathcal{F}\mu)(s) = 0 \quad (s \in i\sigma(A) \cap \mathbb{R}).$$

Then

$$\lim_{t \rightarrow \infty} \|T(t)\hat{\mu}(T)\| = 0.$$

In particular, spectral synthesis is unnecessary in Theorem 1.2 of [1].

4 Tail norm continuity

Recall the critical growth bound $\delta(T)$: it is the infimum of the growth bounds of $T - V$ over all norm-continuous $V : (0, \infty) \rightarrow \mathcal{B}(H)$. The non-analytic Gearhart–Prüss theorem and the standard inequalities for these growth bounds give, on Hilbert space,

$$\delta(T) = s_0^\infty(A).$$

This equality and its use under precisely (1) are recorded in Sections 3–4 of [1], drawing on [4, 5]. A Neumann-series argument applied to the uniformly bounded resolvents in (1) gives

$$\delta(T) = s_0^\infty(A) < 0. \quad (2)$$

Lemma 1 (asymptotic uniform norm continuity). *If a bounded C_0 -semigroup satisfies $\delta(T) < 0$, then for every $\gamma > 0$ there are $r_0 > 0$ and $\eta > 0$ such that*

$$\|T(r+h) - T(r)\| < \gamma$$

whenever $r \geq r_0$, $|h| \leq \eta$, and $r+h \geq 0$.

Proof. Choose a norm-continuous $V : (0, \infty) \rightarrow \mathcal{B}(H)$, $C > 0$, and $c > 0$ such that

$$\|T(r) - V(r)\| \leq Ce^{-cr} \quad (r > 0).$$

For

$$D(r) = \limsup_{h \downarrow 0} \|T(r+h) - T(r)\|,$$

norm continuity of V gives $D(r) \leq 2Ce^{-cr}$, hence $D(r) \rightarrow 0$.

Put $M = \sup_{t \geq 0} \|T(t)\|$. Choose r_1 so large that $MD(r_1) < \gamma/2$. By the definition of the limsup, after decreasing $\eta > 0$ if necessary,

$$\|T(r_1+h) - T(r_1)\| < \gamma/M \quad (0 \leq h \leq \eta).$$

For $r = r_1 + q$, $q \geq 0$, the semigroup law gives

$$\|T(r+h) - T(r)\| \leq M \|T(r_1+h) - T(r_1)\| < \gamma.$$

This handles $h \geq 0$. If $-\eta \leq h < 0$ and $r \geq r_1 + \eta$, apply the positive-increment estimate at the base point $r+h$. Thus $r_0 = r_1 + \eta$ works. $\square$

5 The continuous ultrapower

We isolate the standard ultrapower facts used in the proof. Let $\mathcal{U}$ be a free ultrafilter on $\mathbb{N}$, and let $H_{\mathcal{U}}$ be the Hilbert ultrapower. The operators

$$\tilde{T}(t)[(x_n)] = [(T(t)x_n)]$$

form a bounded semigroup, but need not be strongly continuous on all of $H_{\mathcal{U}}$. For $a \in L^1(\mathbb{R}_+)$, let $\hat{a}(\tilde{T})$ mean the coordinatewise ultrapower of $\hat{a}(T)$, and define

$$H_c = \overline{\text{span}}\{\hat{a}(\tilde{T})\xi : a \in L^1(\mathbb{R}_+), \xi \in H_{\mathcal{U}}\}. \quad (3)$$

Lemma 2. *The restriction $S(t) = \tilde{T}(t)|_{H_c}$ is a bounded C_0 -semigroup. For every $a \in L^1(\mathbb{R}_+)$,*

$$\hat{a}(S) = \hat{a}(\tilde{T})|_{H_c}, \quad \|\hat{a}(S)\| \leq \|\hat{a}(T)\|. \quad (4)$$

Consequently

$$\text{Sp}_u(S) \subset \text{Sp}_u(T). \quad (5)$$

Proof. If $a \in L^1(\mathbb{R}_+)$, translation continuity in L^1 gives

$$\|T(h)\hat{a}(T) - \hat{a}(T)\| \longrightarrow 0 \quad (h \downarrow 0).$$

The same estimate holds in the ultrapower norm. Hence $\tilde{T}$ is strongly continuous on the generators in (3), and therefore on their closed span. Invariance is immediate.

On a generator $\widehat{b}(\widetilde{T})\xi$, Fubini's theorem gives

$$\widehat{a}(S)\widehat{b}(\widetilde{T})\xi = \widehat{a * b}(\widetilde{T})\xi = \widehat{a}(\widetilde{T})\widehat{b}(\widetilde{T})\xi.$$

This proves the first identity in (4) by density. The coordinatewise definition gives the norm inequality.

For completeness, the spectrum of a bounded semigroup representation is characterized by

$$\mathrm{Sp}(T) = \{\chi : |\widehat{a}(\chi)| \leq \|\widehat{a}(T)\| \text{ for every } a \in L^1(\mathbb{R}_+)\}.$$

Thus (4) implies $\mathrm{Sp}(S) \subset \mathrm{Sp}(T)$, and intersecting with the unitary characters proves (5). This is the same spectrum formalism used in [2]. $\square$

We shall also use the standard Hilbert-space unitary-asymptote construction from the proof of Proposition 3.4 of [2].

Lemma 3 (unitary asymptote). *For a bounded C_0 -semigroup S on a Hilbert space G , there exist a Hilbert space K , a strongly continuous unitary group $U = (U(t))_{t \in \mathbb{R}}$ on K , and a bounded operator $\pi : G \rightarrow K$ such that*

$$\pi S(t) = U(t)\pi \quad (t \geq 0), \quad (6)$$

$$\ker \pi = \{x \in G : \|S(t)x\| \rightarrow 0\}, \quad (7)$$

$$\mathrm{Sp}(U) \subset \mathrm{Sp}_u(S). \quad (8)$$

Construction reminder. Apply an invariant Banach limit to $t \mapsto \langle S(t)x, S(t)y \rangle$, quotient by the resulting null space, and complete. This produces an isometric semigroup and the intertwiner. Boundedness of S shows that the null space is exactly the stable vectors: once an orbit has a subnet tending to zero, all of its tail does. The minimal unitary extension gives U . The integrated-form spectrum characterization gives (8); full details are supplied in the cited unitary-asymptote construction [2, proof of Proposition 3.4]. $\square$

6 Proof of Theorem 1

Put

$$M = \sup_{t \geq 0} \|T(t)\|, \quad B = \widehat{\mu}(T).$$

Suppose for contradiction that $\|T(t)B\| \not\rightarrow 0$. Then there are $\varepsilon > 0$, $t_n \rightarrow \infty$, and unit vectors $y_n \in H$ such that

$$\|T(t_n)By_n\| \geq \varepsilon. \quad (9)$$

For $0 \leq q \leq t_n$, the semigroup law and boundedness imply

$$\|T(q)By_n\| \geq \varepsilon/M, \quad (10)$$

because $T(t_n)B = T(t_n - q)T(q)B$.

Fix $\tau > 0$. Since failure implies $B \neq 0$, Lemma 1 supplies $\eta > 0$ and $s_0 > 0$ such that

$$\sup_{0 \leq h \leq \eta} \|T(s + \tau + h) - T(s + \tau)\| < \frac{\varepsilon}{2M\|B\|} \quad (s \geq s_0). \quad (11)$$

Choose a nonnegative $b \in L^1(\mathbb{R}_+)$, supported in $[\tau, \tau + \eta]$, with $\int b = 1$.

Use Lemma 2, put $y = [(y_n)] \in H_{\mathcal{U}}$, and set

$$x = \widehat{b}(\widetilde{T})y = [(\widehat{b}(T)y_n)] \in H_c.$$

The map $q \mapsto T(q)\widehat{b}(T)$ is norm-continuous. Indeed, for positive h , translation continuity of b gives

$$\sup_{q \geq 0} \|T(q+h)\widehat{b}(T) - T(q)\widehat{b}(T)\| \leq M \|T(h)\widehat{b}(T) - \widehat{b}(T)\| \rightarrow 0.$$

It follows that this bounded norm-continuous operator-valued map is Bochner integrable against $|\mu|$. Approximation in $L^1(|\mu|; \mathcal{B}(H))$ by common simple functions gives operator-norm convergence before passing to the ultrapower. Hence, using Fubini,

$$z := \widehat{\mu}(S)x = [(B\widehat{b}(T)y_n)] = [\widehat{\mu * b}(T)y_n] \in H_c. \quad (12)$$

This is the point at which the L^1 smoothing of x is essential; no variation-norm approximation of μ is asserted.

For $s \geq s_0$, equations (10) and (11) give

$$\begin{aligned} \|S(s)z\| &= \lim_{\mathcal{U}} \|T(s)B\widehat{b}(T)y_n\| \\ &= \lim_{\mathcal{U}} \left\| \int_{\tau}^{\tau+\eta} b(q)T(s+q)By_n dq \right\| \\ &\geq \lim_{\mathcal{U}} \|T(s+\tau)By_n\| - \frac{\varepsilon}{2M} \geq \frac{\varepsilon}{2M}. \end{aligned} \quad (13)$$

For the last inequality, the cofinite set on which $t_n \geq s + \tau$ belongs to $\mathcal{U}$.

Apply Lemma 3 to S . From (6), bounded linearity of π , and Bochner integration,

$$\pi z = \int_{\mathbb{R}_+} U(q)\pi x d\mu(q). \quad (14)$$

Under the standard identification of unitary characters with real Fourier parameters, (5) and (8) place the spectral measure of U inside $i\sigma(A) \cap \mathbb{R}$. (Explicitly, the character $t \mapsto e^{-ist}$ has generator $-is$, so $s = i(-is)$.) The spectral theorem and the hypothesis on $\mathcal{F}\mu$ therefore give

$$\int_{\mathbb{R}_+} U(q) d\mu(q) = 0.$$

Thus $\pi z = 0$. By (7), $\|S(s)z\| \rightarrow 0$, contradicting (13). The theorem is proved.

7 Verification, scope, and novelty

- The proof never approximates a singular measure in total variation. The only smoothing is $x = \widehat{b}(\widetilde{T})y$, which makes the operator orbit norm-continuous and justifies (12).
- The ultrapower is restricted to the L^1 -essential subspace (3); strong continuity is not claimed on the full ultrapower.
- No sequential compactness of an operator ball is used. The free ultrafilter preserves the almost-norming sequence directly.
- The resolvent hypothesis is used only to obtain $\delta(T) < 0$, hence Lemma 1. The argument proves the slightly more general statement with $\delta(T) < 0$ in place of (1).
- The result removes spectral synthesis only in the Hilbert-space theorem asked about in Remark 1.3. It makes no analogous assertion for arbitrary Banach spaces.

Novelty check. On 9 August 2026, bounded web/arXiv searches used the exact sentence in Remark 1.3, arXiv:1410.1301, the title and author, and combinations of *bounded Borel measure*, *Hilbert*, *spectral synthesis*, and *Katznelson–Tzafriri*. They found the source, the companion Hilbert paper [2], the original measure theorem [3], and the 2022 survey [6], but no later work claiming this removal. This is a bounded novelty check, not a comprehensive priority search.

Human review recommendation. Check Lemma 1, the integrated-form spectrum inclusion in Lemma 2, the uniform operator-norm approximation behind (12), and the Fourier-sign convention in the final spectral-theorem step. The verifier verdict is *likely valid*.

References

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